

10 Electron-transfer reactions

10.1 Electron-transfer theory

The Hamiltonian written in the diabatic representation is

$$\hat{H} = |0\rangle H_0 \langle 0| + |1\rangle H_1 \langle 1| + |0\rangle \Delta \langle 1| + |1\rangle \Delta \langle 0| \quad (10.1)$$

$$= \sum_j \frac{p_j^2}{2m} + |0\rangle V_0 \langle 0| + |1\rangle V_1 \langle 1| + |0\rangle \Delta \langle 1| + |1\rangle \Delta \langle 0| \quad (10.2)$$

where $|0\rangle$ and $|1\rangle$ are the reactant and product electronic states.

We identify the important operators:

- projection onto reactants is $|0\rangle\langle 0|$
- projection onto products is $|1\rangle\langle 1|$
- flux is $\hat{F} = \frac{i}{\hbar} [\hat{H}, |1\rangle\langle 1|] = \frac{i}{\hbar} (\hat{H} |1\rangle\langle 1| - |1\rangle\langle 1| \hat{H}) = \frac{i}{\hbar} \Delta (|0\rangle\langle 1| - |1\rangle\langle 0|)$

which you check are all Hermitian.

To lowest order in Δ , the reactant partition function is

$$Z_0 = \text{Tr} [e^{-\beta \hat{H}} |0\rangle\langle 0|] \sim \text{Tr}[e^{-\beta \hat{H}_0}]$$

and the flux-flux correlation function is

$$\begin{aligned} C_{\text{ff}} &= \text{Tr} [e^{-\beta \hat{H}} \hat{F} e^{i\hat{H}t/\hbar} \hat{F} e^{-i\hat{H}t/\hbar}] \\ &= \left(\frac{i}{\hbar} \Delta \right)^2 \text{Tr} [e^{-\beta \hat{H}} |0\rangle\langle 1| e^{i\hat{H}t/\hbar} |0\rangle\langle 1| e^{-i\hat{H}t/\hbar}] - \left(\frac{i}{\hbar} \Delta \right)^2 \text{Tr} [e^{-\beta \hat{H}} |0\rangle\langle 1| e^{i\hat{H}t/\hbar} |1\rangle\langle 0| e^{-i\hat{H}t/\hbar}] \\ &\quad + \text{“other way around”} \\ &\sim \frac{\Delta^2}{\hbar^2} \text{Tr} [e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} e^{-i\hat{H}_0 t/\hbar}] + \text{“other way around”} \end{aligned}$$

We can thus compute the rate using

$$\begin{aligned} kZ_0 &= \int_{-\infty}^{\infty} \frac{1}{2} C_{\text{ff}}(t) dt \\ &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \text{Tr} [e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} e^{-i\hat{H}_0 t/\hbar}] dt \end{aligned}$$

where we have used the fact that both terms in the correlation function integrate to the same number, so we only need one (multiplied by 2).

It is possible to show that this formulation for the rate is equivalent to Fermi's golden rule given earlier in the course. We expand the trace using the eigenstate basis, defined such that

$\hat{H}_0 |\psi_0^\mu\rangle = E_0^\mu |\psi_0^\mu\rangle$ and $\hat{H}_1 |\psi_1^\nu\rangle = E_1^\nu |\psi_1^\nu\rangle$ and insert complete sets of states $1 = \sum_\mu |\psi_0^\mu\rangle\langle\psi_0^\mu|$ and $1 = \sum_\nu |\psi_1^\nu\rangle\langle\psi_1^\nu|$. This gives

$$\begin{aligned}
 kZ_0 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \text{Tr} \left[e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} e^{-i\hat{H}_0 t/\hbar} \right] dt \\
 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \sum_\mu \sum_\nu \left\langle \psi_0^\mu \left| e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} \right| \psi_1^\nu \right\rangle \left\langle \psi_1^\nu \left| e^{-i\hat{H}_0 t/\hbar} \right| \psi_0^\mu \right\rangle dt \\
 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \sum_\mu \sum_\nu e^{-\beta E_0^\mu} \langle \psi_0^\mu | \psi_1^\nu \rangle e^{iE_1^\nu t/\hbar} \langle \psi_1^\nu | \psi_0^\mu \rangle e^{-iE_0^\mu t/\hbar} dt \\
 &= \frac{\Delta^2}{\hbar^2} \sum_\mu e^{-\beta E_0^\mu} \sum_\nu |\langle \psi_0^\mu | \psi_1^\nu \rangle|^2 \int_{-\infty}^{\infty} e^{i(E_1^\nu - E_0^\mu)t/\hbar} dt \\
 &= \frac{2\pi\Delta^2}{\hbar} \sum_\mu e^{-\beta E_0^\mu} \sum_\nu \underbrace{|\langle \psi_0^\mu | \psi_1^\nu \rangle|^2}_{\text{Franck-Condon factor}} \delta(E_0^\mu - E_1^\nu)
 \end{aligned}$$

which is identical to the previous formula if we identify the couplings $\Delta_{\mu\nu} = \Delta \langle \psi_0^\mu | \psi_1^\nu \rangle$.

Now we take the classical limit by converting quantum trace into phase-space averages.

$$Z_0 = \text{Tr}[e^{-\beta \hat{H}_0}] \rightarrow (2\pi\hbar)^{-f} \iint e^{-\beta H_0} dp dq$$

$$\begin{aligned}
 kZ_0 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \text{Tr} \left[e^{-\beta \hat{H}_0} e^{i\hat{H}_0 t/\hbar} e^{-i\hat{H}_1 t/\hbar} \right] dt \\
 &\rightarrow (2\pi\hbar)^{-f} \frac{\Delta^2}{\hbar^2} \iiint e^{-\beta H_0} e^{iH_0 t/\hbar} e^{-iH_1 t/\hbar} dp dq dt \\
 &= (2\pi\hbar)^{-f} \frac{\Delta^2}{\hbar^2} \iint e^{-\beta H_0} \int e^{i(H_0 - H_1)t/\hbar} dt dp dq \\
 &= (2\pi\hbar)^{-f} \frac{2\pi\Delta^2}{\hbar} \iint e^{-\beta H_0} \delta(H_0 - H_1) dp dq \\
 &= \frac{2\pi\Delta^2}{\hbar} \langle \delta(H_0 - H_1) \rangle_0 \\
 &= \frac{2\pi\Delta^2}{\hbar} \langle \delta(V_0 - V_1) \rangle_0
 \end{aligned}$$

where the final expressions are classical averages over the Boltzmann distribution of the reactant. These can be computed using molecular dynamics simulations.

10.2 Marcus theory

In certain cases, it is appropriate to assume that the Hamiltonians can be represented in harmonic system-bath form.

$$H_0 = \frac{p^2}{2M} + \sum_j \frac{p_j^2}{2m} + \frac{1}{2}M\Omega^2(x + \xi)^2 + \sum_j \frac{1}{2}m\omega_j^2 \left(q_j + \frac{c_j}{m\omega_j^2}x \right)^2$$

$$H_1 = \frac{p^2}{2M} + \sum_j \frac{p_j^2}{2m} + \frac{1}{2}M\Omega^2(x - \xi)^2 + \sum_j \frac{1}{2}m\omega_j^2 \left(q_j - \frac{c_j}{m\omega_j^2}x \right)^2 - \varepsilon$$

We define the reorganization energies of reactant and product:

$$\Lambda_0 = V_0(x_{\min}^{(1)}) - V_0(x_{\min}^{(0)}) = 2M\Omega^2\xi^2$$

$$\Lambda_1 = V_1(x_{\min}^{(0)}) - V_1(x_{\min}^{(1)}) = 2M\Omega^2\xi^2$$

which are seen to be equal to each other.

Using this harmonic assumption, we can compute the rate (which will turn out to be Marcus theory)

$$\begin{aligned} k_{\text{MT}} &= \frac{2\pi\Delta^2}{\hbar} \frac{\int e^{-\beta V_0} \delta(V_0 - V_1) dx d\mathbf{q}}{\int e^{-\beta V_0} dx d\mathbf{q}} \\ &= \frac{2\pi\Delta^2}{\hbar} \frac{\int e^{-\beta V_0} \delta(2M\Omega^2\xi x + \varepsilon) dx d\mathbf{q}}{\int e^{-\beta V_0} dx d\mathbf{q}} \\ &= \frac{2\pi\Delta^2}{\hbar} \frac{\int e^{-\beta V_0(x)} \delta(2M\Omega^2\xi x + \varepsilon) dx}{\int e^{-\beta M\Omega^2(x+\xi)^2/2} dx} \\ &= \frac{2\pi\Delta^2}{\hbar} \frac{e^{-\beta V_0(x^\ddagger)} \frac{1}{2M\Omega^2\xi}}{\sqrt{2\pi/\beta M\Omega^2}} \\ &= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{2M\Omega^2\xi^2}} e^{-\beta V^\ddagger} \\ &= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{\Lambda}} e^{-\beta V^\ddagger} \end{aligned}$$

where $x^\ddagger = -\varepsilon/2M\Omega^2\xi$ and the activation energy is

$$\begin{aligned} V^\ddagger &= \frac{1}{2}M\Omega^2(x^\ddagger + \xi)^2 \\ &= \frac{1}{2}M\Omega^2 \left(-\frac{\varepsilon}{2M\Omega^2\xi} + \xi \right)^2 \\ &= (2M\Omega^2\xi^2 - \varepsilon)^2 / 8M\Omega^2\xi^2 \\ &= (\Lambda - \varepsilon)^2 / 4\Lambda \end{aligned}$$

such that the Marcus theory rate can be written in its usual form

$$k_{\text{MT}} = \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{\Lambda}} e^{-\beta(\Lambda - \varepsilon)^2 / 4\Lambda}$$

This equation predicts that the rate is slow when ε is small, that the rate increases as ε approaches Λ at which it reaches a maximum where the reaction is activationless. Then as ε increases past Λ , the reaction is in the inverted regime and the rate decreases again. In reality (as shown in experiment and simulations taking account of quantum effects), tunnelling causes the rate to increase substantially in the inverted regime, skewing the Marcus turnover curve.

10.3 Adiabatic dynamics: Quantum adiabatic theorem

We consider a quantum system with discrete level structure and a Hamiltonian given by:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t). \quad (10.3)$$

The matrix representation of $\hat{H}$ in the basis of eigenstates of $\hat{H}_0$ is:

$$H(t) = \begin{pmatrix} E_a & V_{ab}(t) & \dots \\ V_{ba}(t) & E_b & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

We seek a solution of the corresponding TDSE, which can be written as

$$\psi(t) = \begin{pmatrix} \psi_a(t) \\ \psi_b(t) \\ \vdots \end{pmatrix}.$$

One can define a unitary transformation that diagonalizes the instantaneous Hamiltonian $H(t)$:

$$\begin{aligned} S^\dagger(t)H(t)S(t) &= D(t) \\ S^\dagger(t)\psi(t) &= \tilde{\psi}(t), \end{aligned} \quad (10.4)$$

where $D(t)$ is a diagonal matrix and $S^\dagger(t) = S^{-1}(t)$.

The TDSE can be written as

$$i\hbar \frac{\partial}{\partial t} (S(t)\tilde{\psi}(t)) = i\hbar \left(S(t) \frac{\partial \tilde{\psi}(t)}{\partial t} + \frac{\partial S(t)}{\partial t} \tilde{\psi}(t) \right) = H(t)S(t)\tilde{\psi}(t). \quad (10.5)$$

Pre-multiplying both sides by $S^\dagger(t)$ and rearranging gives

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}(t) = D(t)\tilde{\psi}(t) - i\hbar S^\dagger(t) \frac{\partial S(t)}{\partial t} \tilde{\psi}(t). \quad (10.6)$$

N.B. $S^\dagger(t)S(t) = I$ so $0 = \frac{\partial}{\partial t} S^\dagger(t)S(t) = \frac{\partial S^\dagger(t)}{\partial t} S(t) + S^\dagger(t) \frac{\partial S(t)}{\partial t}$ so the last line can be alternatively written as

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}(t) = D(t)\tilde{\psi}(t) + i\hbar \frac{\partial S^\dagger(t)}{\partial t} S(t)\tilde{\psi}(t).$$

If $H(t)$ is slowly varying, so will be $S(t)$, and in this case $\frac{\partial S(t)}{\partial t}$ may be neglected. This is called the *adiabatic approximation*.

Adiabatic theorem: If $\psi(0)$ is an eigenfunction of $H(0)$ and $H(t)$ is a slowly-varying function of time, then $\psi(t)$ will evolve in such a way as to remain an eigenfunction of $H(t)$ for all time, in the limit of infinitely-slowly-varying $H(t)$. (see A.Messiah Quantum Mechanics, North Holland Publishing Company, 1967 for the detailed derivation).

10.4 Landau–Zener theory

According to Landau–Zener theory, the transition probability between two diabatic states is

$$P_{1 \leftarrow 0} = 1 - \exp \left[-\frac{2\pi\Delta^2}{\hbar \left| \frac{\partial}{\partial t} (E_0(x) - E_1(x)) \right|} \right]$$

This formula is an exact solution to the time-dependent dynamics of a two-level system, in which the energies change linearly with time. It is however used as an approximation to the coupled problem of a nonadiabatic transition. In this case, we use $E_n(x(t)) = V^\ddagger + vF_nt$, where v is the velocity of the particle and F_n the force in state n .

$$P_{1 \leftarrow 0} = 1 - \exp \left[-\frac{2\pi\Delta^2}{\hbar v |F_0 - F_1|} \right]$$

In the weak-coupling limit (or equivalently for fast moving particles or very sharp barriers)

$$P_{1 \leftarrow 0} \rightarrow \frac{2\pi\Delta^2}{\hbar v |F_0 - F_1|}$$

Whereas in the strong-coupling limit:

$$P_{1 \leftarrow 0} \rightarrow 1$$

This strong-coupling limit implies that a diabatic transition is inevitable, or in other words that the adiabatic transition is very unlikely to happen. This means that the adiabatic (Born–Oppenheimer) approximation is valid and is in agreement with what we noticed in previous lectures: that the adiabatic approximation is good for slow moving particles and also for large energy separation between the adiabatic states, which is a consequence of large Δ .

Compare these findings with those of the dynamics of a time-independent two-level system studied earlier in the course. There, we found that the population transfer was only large if the initial and final state were degenerate. In Landau–Zener theory, the transfer occurs mostly near the point when the two states are close together, as you will show in the homework exercise.

10.5 Nonadiabatic transition-state theory

Here we will take a one-dimensional system. Coupling to other modes can be included using the techniques of the previous chapter. First we will consider the TST rate in the adiabatic limit, which

can be written

$$\begin{aligned}
 k_{\text{TST}} &= \frac{(2\pi\hbar)^{-1} \iint e^{-\beta H_0} \delta(x - x^\ddagger) \frac{p}{m} \Theta(p) \, dx \, dp}{(2\pi\hbar)^{-1} \iint e^{-\beta H_0} \, dx \, dp} \\
 &= \frac{\beta\Omega}{2\pi} \iint e^{-\beta H_0} \delta(x - x^\ddagger) \frac{p}{m} \Theta(p) \, dx \, dp \\
 &= \frac{\Omega}{2\pi} \int e^{-\beta V_0} \delta(x - x^\ddagger) \, dx \\
 &= \frac{\Omega}{2\pi} e^{-\beta V^\ddagger}
 \end{aligned}$$

where we used the classical vibrational partition function $Z_0 = \frac{1}{\beta\hbar\Omega}$.

Let's propose a nonadiabatic transition-state theory as

$$k_{\text{NTST}} = \kappa k_{\text{TST}} = \kappa \frac{\Omega}{2\pi} e^{-\beta V^\ddagger}$$

where κ is a transmission coefficient which includes all nonadiabatic effects.

To derive a formula for κ , we start again from the flux-side correlation function formalism but also include an extra weight $T(v)$, which is the transmission probability at a given velocity.

$$\begin{aligned}
 k_{\text{NTST}} &= \frac{\beta\Omega}{2\pi} \iint e^{-\beta H_0} \delta(x - x^\ddagger) \frac{p}{M} T(p/M) \Theta(p) \, dx \, dp \\
 &= \frac{\beta\Omega}{2\pi} \iint e^{-\beta p^2/2M - \beta V_0} \delta(x - x^\ddagger) \frac{p}{M} T(p/M) \Theta(p) \, dx \, dp \\
 &= \frac{\beta M \Omega}{2\pi} \iint e^{-\beta M v^2/2 - \beta V_0} \delta(x - x^\ddagger) v T(v) \Theta(v) \, dx \, dv \\
 &= \frac{\beta M \Omega}{2\pi} \left[\int_0^\infty v e^{-\beta M v^2/2} T(v) \, dv \right] e^{-\beta V^\ddagger} \\
 &= \kappa \frac{\Omega}{2\pi} e^{-\beta V^\ddagger}
 \end{aligned}$$

so

$$\kappa = \beta M \int_0^\infty v e^{-\beta M v^2/2} T(v) \, dv$$

The transmission probability can be obtained using an argument based on Landau–Zener (LZ) theory. LZ theory tells us that a diabatic-state transition will occur with probability P . Thus we take $T(v) = P + \dots$. However, if the reaction remains on the same diabatic state (which has probability $1 - P$), it will eventually reach a region with strong forces pushing it backwards so it returns to pass the crossing point again. Here it again has a probability of P for changing diabatic state and a probability of $1 - P$ of returning to the reactant well. By following the motion of the

particle in this way, we can deduce that

$$\begin{aligned}
 T(v) &= P + (1 - P)P(1 - P) + (1 - P)P^3(1 - P) + \dots \\
 &= P + (1 - P)P \sum_n (P^2)^n (1 - P) \\
 &= P + (1 - P)P \frac{1 - P}{1 - P^2} \\
 &= P + (1 - P)P \frac{1}{1 + P} \\
 &= \frac{P(1 + P) + (1 - P)P}{1 + P} \\
 &= \frac{2P}{1 + P}
 \end{aligned}$$

We consider the limiting case of strong coupling, where $P \rightarrow 1$ and therefore $T(v) \rightarrow 1$ such that

$$\kappa \rightarrow \beta M \int_0^\infty v e^{-\beta M v^2/2} dv = 1$$

As expected in this strong-coupling limit, we recover the adiabatic TST result.

$$k_{\text{NTST}} \rightarrow k_{\text{TST}}$$

In the weak-coupling limit we have small $P \rightarrow \frac{2\pi\Delta^2}{\hbar v|F_0 - F_1|} \ll 1$ and thus $T(v) \rightarrow 2P$

$$\begin{aligned}
 \kappa &\rightarrow \beta M \int_0^\infty v e^{-\beta M v^2/2} \frac{4\pi\Delta^2}{\hbar v|F_0 - F_1|} dv \\
 &= \frac{1}{2}\beta M \frac{4\pi\Delta^2}{\hbar|F_0 - F_1|} \sqrt{\frac{2\pi}{\beta M}} = \frac{2\pi\Delta^2}{\hbar|F_0 - F_1|} \sqrt{2\pi\beta M}
 \end{aligned}$$

so

$$k_{\text{NTST}} = \frac{\Omega}{2\pi} \frac{2\pi\Delta^2}{\hbar|F_0 - F_1|} \sqrt{2\pi\beta M} e^{-\beta V^\ddagger} = \frac{\Delta^2}{\hbar|F_0 - F_1|} \Omega \sqrt{2\pi\beta M} e^{-\beta V^\ddagger}$$

which using $|F_0 - F_1| = |M\Omega^2(x + \xi) - M\Omega^2(x - \xi)| = 2M\Omega^2\xi$ can be shown to be equal to Marcus theory:

$$\begin{aligned}
 k_{\text{NTST}} &= \frac{\Delta^2}{\hbar} \frac{\sqrt{2\pi\beta M}}{2M\Omega\xi} e^{-\beta V^\ddagger} \\
 &= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{2M\Omega^2\xi^2}} e^{-\beta V^\ddagger} \\
 &= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{\Lambda}} e^{-\beta V^\ddagger}
 \end{aligned}$$

Thus in the weak-coupling limit, we recover the golden-rule result and in the strong-coupling limit the adiabatic result. This shows that our NTST approach has the correct behaviour.

10.6 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Electron-transfer theory

- Chandler, Electron transfer in water and other polar environments, how it happens *from: Classical and Quantum Dynamics in Condensed Phase Simulations*, edited by B. J. Berne, G. Ciccotti and D. F. Coker (World Scientific, Singapore, 1998), pgs. 25-49.
<http://stage.cchem.berkeley.edu/~dcgrp/Pubs/DC154.pdf>

Landau-Zener theory

- C. Zener, “Non-adiabatic crossing of energy levels”, *Proc. R. Soc. Lond. A*, **137**, 696 (1932).
<http://dx.doi.org/10.1098/rspa.1932.0165>